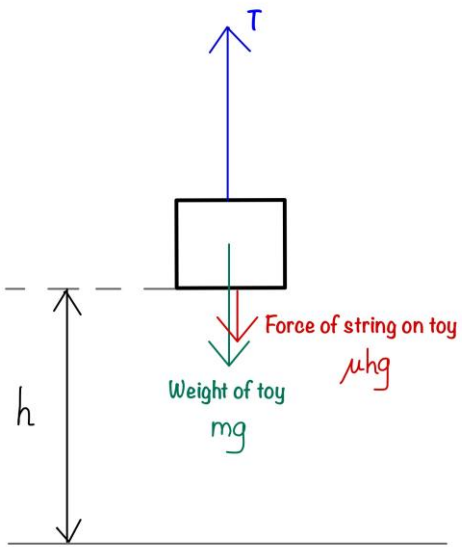
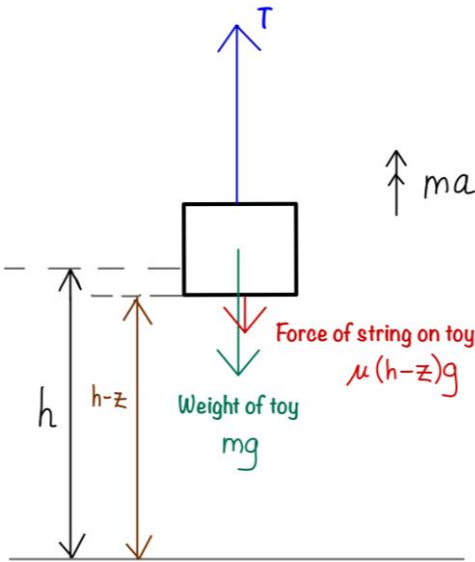
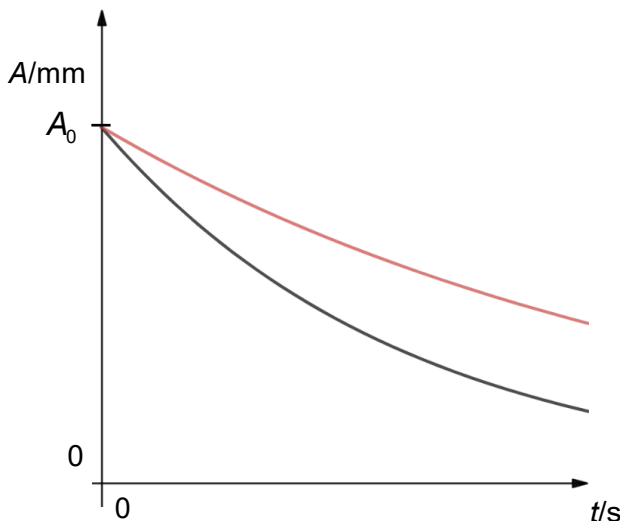
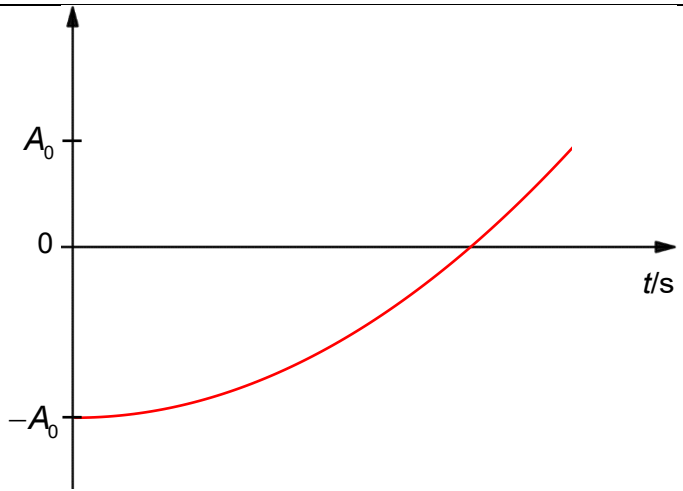


|     |                                                                                                                                                                                                                                                                                                                       |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5   |                                                                                                                                                                                                                                                                                                                       |
| (a) |                                                                                                                                                                                                                                     |
|     | <p>In vertical equilibrium, the upward force (by balloon) is equal to the total downward forces.</p> <p>The <u>downward forces</u> on the toy are the <u>weight of the toy</u> and the tension due to the string on the toy. The tension is equal to the <u>weight of</u> the vertical part of <u>the string</u>.</p> |
| (b) |                                                                                                                                                                                                                                    |
|     | Define upwards to be positive.                                                                                                                                                                                                                                                                                        |
|     | <p>Net force acting on the toy at the instant the toy is displaced downwards,</p> $F_{\text{net}} = T - mg - \mu(h - z)g$ <p>Applying Newton's second law,</p> $ma = T - mg - \mu(h - z)g$                                                                                                                            |
|     | <p>Substituting the expression for <math>T</math> from (a),</p> $ma = mg + \mu gh - mg - \mu(h - z)g$ $a = \frac{\mu g}{m} z$                                                                                                                                                                                         |
|     | When the toy is displaced downwards, it experiences a net force and hence acceleration upwards.                                                                                                                                                                                                                       |

|          |                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Therefore, <u>acceleration is in opposite direction to displacement.</u>                                                                                                                            |
|          | $a = -\left(\frac{\mu g}{m}\right)z$                                                                                                                                                                |
|          | OR                                                                                                                                                                                                  |
|          | Define downwards to be positive.                                                                                                                                                                    |
|          | Net force acting on the toy at the instant the toy is displaced downwards,<br>$F_{\text{net}} = mg + \mu(h - z)g - T$<br>Applying Newton's second law,<br>$ma = mg + \mu(h - z)g - T$               |
|          | Substituting the expression for $T$ from (a),<br>$ma = mg + \mu(h - z)g - mg + \mu gh$                                                                                                              |
|          | $a = -\frac{\mu g}{m}z$                                                                                                                                                                             |
| (c)      | $\omega^2 = \frac{\mu g}{m}$<br>$T = 2\pi\sqrt{\frac{m}{\mu g}}$                                                                                                                                    |
|          | $T = 2\pi\sqrt{\frac{50}{5.8 \times 9.81}}$                                                                                                                                                         |
|          | $T = 5.89 \text{ s (3sf)}$                                                                                                                                                                          |
| (d)(i)   | $A$ <u>decreases exponentially</u> over time.                                                                                                                                                       |
| (d)(ii)  | A smaller base surface area means the extent of damping is lesser.<br><div style="text-align: center;">  </div> |
| (d)(iii) | Each time the system is displaced downwards, the <u>string collides inelastically</u> with the <u>ground</u> .                                                                                      |
|          | The system needs to <u>do work against the force of the ground on the system</u> resulting in further energy dissipated from the system.                                                            |
|          | OR                                                                                                                                                                                                  |
|          | Air currents may <u>drag the system across horizontally</u> .                                                                                                                                       |
|          | The system needs to <u>do work against frictional force</u> resulting in further energy dissipated from the system.                                                                                 |

|         |                                                                                                                                                                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | OR                                                                                                                                                                                                                                                                      |
|         | There is <u>frictional force</u> between the <u>ground</u> and the <u>string</u> .                                                                                                                                                                                      |
|         | The system needs to <u>do work against frictional force</u> resulting in further energy dissipated from the system.                                                                                                                                                     |
| (e)(i)  |                                                                                                                                                                                       |
| (e)(ii) | The only forces acting on the system are the <u>upthrust on the balloon</u> and the <u>weight</u> of the system. These are constant. Hence, the <u>net upwards force</u> is <u>constant</u> and the system will move <u>upwards</u> with <u>constant acceleration</u> . |
| 6       |                                                                                                                                                                                                                                                                         |